

CRYPTOLOGY

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Topic: CO1

Security Features in Real-World Applications

1. Identify any five real-world applications or online services that involve communication, authentication, or transactions. For each application, investigate and identify:

Application	Security feature	Security Mechanism	Protocol/ Technology	How it provides security
Gmail	Authentication	Password and multi-factor authentication	OAuth 2.0, TLS	It make sure that only the actual user can login to the account
Zoom	Confidentiality	Encryption	TLS, AES encryption	It keeps the meeting private so that other people cannot see or get the access
Dropbox	Authorization	Access control	OAuth 2.0, TLS	Only people who are having the permission can access and edit the file
Amazon	Integrity	Data protection	TLS, HTTPS	it protects the customer and payment details from being stolen or changed
Microsoft teams	Availability	Redundant infrastructure	Cloud computing, load balancing	The service will be available even if the server is having the problem

2. For any five real-world applications, investigate the security controls used to protect users and data. Classify the controls based on whether they provide Confidentiality, Integrity, or Availability, and determine how these controls help in preventing, detecting, and recovering from security attacks. Mention the mechanisms and protocols involved.

Application	Security Control	CIA	Prevention	Detection	Recovery	Mechanism and Protocol
Gmail	Password and multi-factor authentication	Confidentiality	MFA adds an extra step to make sure only the account owner can log in.	Its alerts the user when there is a login they don't recognize	The user can change the passwords and get the account back if there is a problem	MFA, OAuth 2.0, TLS
Zoom	Encryption and Access Control	Confidentiality	It keeps the meeting private other people	It can also detect if someone tries to join without permission	The host can end the meeting and stop unwanted people from joining	AES, TLS, encryption
Dropbox	File Permissions and Encryption	Confidentiality, integrity	Only people who have the permission can open or edit the file	Its helps to find out if someone makes unwanted changes to a file	If something goes wrong an older version of the file can be restored	AES, TLS, OAuth 2.0
Amazon	Transaction Authentication and Encryption	Confidentiality, integrity	It keeps customer and payment details safe while making a payment	It can spot unusual or suspicious transactions	If something goes wrong, the account can be secured and the transaction can be reported	HTTPS, TLS, MFA, encryption

Microsoft teams	Access Control and Encryption	Confidentiality, availability	It makes sure only allowed people can join or access the information	It helps find suspicious activity or problem with the service	If service goes down backup systems help bring it back	TLS, encryption, MFA, cloud infrastructure
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